

Convexity of the ordinary infinite-dimensional C -numerical range for positive C with finite kernel

Candidate partial result for arXiv:2009.01300
Lane 12, agent `agent_lane_12`, model GPT5.6

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Status: partial result, likely valid; human verification requested. The proof gives a complete positive answer when $C \geq 0$ is trace class with $\dim \ker C < \infty$, for every bounded A . The general selfadjoint problem, and the positive infinite-rank/infinite-kernel case for arbitrary A , remain open in this packet.

1 Source question and result

Let $\mathcal{H}$ be an infinite-dimensional separable complex Hilbert space. For $C \in \mathcal{L}^1(\mathcal{H})$ and $A \in \mathcal{B}(\mathcal{H})$, the ordinary C -numerical range is

$$W_C(A) := \{\operatorname{Tr}(UCU^*A) : U \in \mathcal{U}(\mathcal{H})\}.$$

Dirr and vom Ende asked whether $W_C(A)$ is convex whenever C is selfadjoint [1, Open Problem (b)]. Loreaux and Patnaik restated the question and proved convexity for their orbit-closed modification, together with ordinary convexity under special hypotheses [2]. The question is recorded in the abstract of arXiv:2009.01300, reproduced below.

Transcription of the question. For selfadjoint trace-class C and bounded A , is $W_C(A)$ convex without taking its closure?

Our result settles the finite-kernel positive subcase without any hypothesis on A .

Theorem 1 (Finite-kernel positive case). *Let $C \in \mathcal{L}^1(\mathcal{H})$ be positive and suppose $\dim \ker C < \infty$. Then $W_C(A)$ is convex for every $A \in \mathcal{B}(\mathcal{H})$.*

Because $\mathcal{H}$ is infinite-dimensional, such a C has infinite rank. The finite-rank case is already known from Westwick’s theorem [3].

2 Proof intuition

There are two different geometric regimes. First, every point in the planar interior of $\overline{W_C(A)}$ is already in $W_C(A)$. To see this, surround the point by three orbit values, approximate the corresponding three orbit operators so that they share one common infinite-dimensional tail, and vary only a common finite-dimensional head. The finite-dimensional C -numerical range of that head is convex, so it fills the triangle exactly.

Only boundary segments remain. After rotating a supporting line, points on such a segment are ordinary maximizers of $X \mapsto \operatorname{Tr}(XH)$ for a selfadjoint H . The equality analysis of Loreaux–Patnaik

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ARTICLE HISTORY

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ABSTRACT

We introduce and investigate the orbit-closed C -numerical range, a natural modification of the C -numerical range of an operator introduced for C trace-class by Dirr and vom Ende. Our orbit-closed C -numerical range is a conservative modification of theirs because these two sets have the same closure and even coincide when C is finite rank. Since Dirr and vom Ende's results concerning the C -numerical range depend only on its closure, our orbit-closed C -numerical range inherits these properties, but we also establish more.

For C selfadjoint, Dirr and vom Ende were only able to prove that the closure of their C -numerical range is convex, and asked whether it is convex without taking the closure. We establish the convexity of the orbit-closed C -numerical range for selfadjoint C without taking the closure by providing a characterization in terms of majorization, unlocking the door to a plethora of results which generalize properties of the C -numerical range known in finite dimensions or when C has finite rank. Under rather special hypotheses on the operators, we also show the C -numerical range is convex, thereby providing a partial answer to the question posed by Dirr and vom Ende.

KEYWORDS

numerical range, C -numerical range, convex, trace-class, Toeplitz–Hausdorff Theorem, unitary orbit, Hausdorff distance, essential spectrum

AMS CLASSIFICATION

Primary 47A12, 47B15; Secondary 52A10, 52A40, 26D15.

1. Introduction

Herein we let $\mathcal{H}$ denote a separable complex Hilbert space and $B(\mathcal{H})$ the collection of all bounded linear operators on $\mathcal{H}$. For $A \in B(\mathcal{H})$, the *numerical range* $W(A)$ is

Figure 1: Full-width crop of PDF page 1 of arXiv:2009.01300. The abstract says that Dirr–vom Ende asked whether the ordinary C -numerical range is convex without taking closure, and describes the paper's result as only a partial answer for the ordinary range.

forces every maximizer to put the same successive eigenvalue blocks of C into the spectral blocks of H above the essential spectral maximum. It vanishes below that maximum. The finite-kernel assumption fixes the kernel multiplicity of the sole residual block at the threshold. Hence any two ordinary maximizers are conjugate by a unitary commuting with H . A logarithm path in that commutant stays on the supporting face, and the intermediate value theorem fills the segment.

3 Interior points

The first lemma is stated for arbitrary selfadjoint trace-class C .

Lemma 2 (Common-tail approximation). *Let $C = C^* \in \mathcal{L}^1(\mathcal{H})$ and let $X_j = U_j C U_j^*$, $1 \leq j \leq r$, be a finite family in its unitary orbit. For every $\varepsilon > 0$ there are a finite-dimensional subspace $M \subset \mathcal{H}$, a single trace-class operator D on $M^\perp$, and orbit elements*

$$Y_j = B_j \oplus D \quad (\mathcal{H} = M \oplus M^\perp)$$

such that $\|X_j - Y_j\|_1 < \varepsilon$ for all j , and all B_j belong to one finite-dimensional selfadjoint unitary orbit in $\mathcal{B}(M)$.

Proof. Choose a finite-rank spectral truncation C_N of C such that $2\|C - C_N\|_1 < \varepsilon$, and let E_N be its supporting finite-dimensional spectral subspace. Choose a finite-dimensional M containing

all $U_j E_N$ and put $d = \dim M$. There is a C -invariant d -dimensional subspace $E_d \supset E_N$: extend by further nonzero eigenvectors, and then by kernel vectors if necessary. Choose an eigenbasis $e_1, \dots, e_d$ of E_d whose initial vectors support C_N .

For each j , extend the prescribed orthonormal family obtained from those initial vectors under U_j to a basis $f_{j,1}, \dots, f_{j,d}$ of M . Fix a unitary $R : E_d^\perp \rightarrow M^\perp$, and define a unitary V_j by

$$V_j e_n = f_{j,n} \quad (1 \leq n \leq d), \quad V_j|_{E_d^\perp} = R.$$

Set $Y_j = V_j C V_j^*$. The C_N parts of X_j and Y_j coincide, whence

$$\|X_j - Y_j\|_1 \leq \|U_j(C - C_N)U_j^*\|_1 + \|V_j(C - C_N)V_j^*\|_1 = 2\|C - C_N\|_1 < \varepsilon.$$

Relative to $M \oplus M^\perp$, the head B_j has the eigenvalue list of $C|_{E_d}$, independent of j , and the tail is the common operator $D = R(C|_{E_d^\perp})R^*$. $\square$

Lemma 3 (Interior filling). *If $C = C^* \in \mathcal{L}^1(\mathcal{H})$ and $A \in \mathcal{B}(\mathcal{H})$, then*

$$\text{int } \overline{W_C(A)} \subseteq W_C(A),$$

where the interior is the ordinary planar interior.

Proof. Loreaux–Patnaik prove that the orbit-closed range is convex and that $W_C(A)$ is dense in it [2, Theorem 3.4 and Corollary 4.6]. Thus $K := \overline{W_C(A)}$ is convex. Let $z \in \text{int } K$. Choose three values $z_j = \text{Tr}(X_j A)$ in $W_C(A)$ whose triangle contains z in its strict interior. This is possible by first taking a small triangle around z inside K , then perturbing its vertices into the dense subset $W_C(A)$.

Apply Lemma 2 with error small enough that $y_j := \text{Tr}(Y_j A)$ still have a triangle containing z ; indeed $|y_j - z_j| \leq \|Y_j - X_j\|_1 \|A\|$. Let $A_M = P_M A|_M$. Keeping the common tail D fixed and varying the finite head through its unitary orbit gives exactly the subset

$$\text{Tr}(D P_{M^\perp} A|_{M^\perp}) + W_{C_M}(A_M) \subseteq W_C(A), \quad (3.1)$$

where $C_M = C|_{E_d}$ transported to M . Off-diagonal blocks of A make no contribution to the trace. The set in (3.1) is convex by the finite-dimensional C -numerical-range theorem for selfadjoint C_M [3, 4]. It contains all three y_j , hence their triangle and therefore z . $\square$

4 Supporting faces

We now assume $C \geq 0$ and put $k = \dim \ker C < \infty$. Denote by $\mathcal{O}(C)$ the trace-norm closure of the unitary orbit.

Lemma 4 (One commutant orbit of ordinary maximizers). *Let $H = H^* \in \mathcal{B}(\mathcal{H})$. If $X_1, X_2 \in \mathcal{U}(C)$ both maximize $X \mapsto \text{Tr}(XH)$ over $\mathcal{U}(C)$, then there is a unitary V commuting with H such that $X_2 = V X_1 V^*$.*

Proof. Let $m = \max \sigma_{\text{ess}}(H)$ and $P = \chi_{[m, \infty)}(H)$. Density of the ordinary range in the orbit-closed range makes the two suprema equal. Therefore the maximizer-support theorem [2, Theorem 5.7] applies and gives

$$X_j = P X_j P. \quad (4.1)$$

Since X_j is unitarily equivalent to C , its kernel has dimension k . Thus $P^\perp \mathcal{H} \subseteq \ker X_j$, so

$$q := \dim P^\perp \leq k < \infty. \quad (4.2)$$

List the distinct eigenvalues of H strictly above m in decreasing order, and let P_ℓ be the associated finite-dimensional spectral projections. The block-decomposition equality theorem [2, Proposition 5.12] says that each X_j commutes with every P_ℓ , and its compression to $P_\ell \mathcal{H}$ is unitarily equivalent to the same successive finite block of the decreasing positive eigenvalues of C .

Suppose first that there are infinitely many such blocks. Their cumulative dimensions tend to infinity, so the prescribed blocks exhaust every nonzero eigenvalue of C . Their traces therefore sum to $\operatorname{Tr} C = \operatorname{Tr} X_j$. Positivity forces X_j to vanish on the orthogonal complement of their direct sum. The two compressions are unitarily conjugate on each block and both are zero on the complement.

Suppose instead that there are only finitely many blocks strictly above m . Proposition 5.12 places the remaining nonzero eigenvalue tail of C on the threshold space $P_0 \mathcal{H}$, where $P_0 = \chi_{\{m\}}(H)$, while (4.1) makes X_j zero on $P^\perp \mathcal{H}$. The blocks strictly above m contain only positive eigenvalues and have no kernel. Hence

$$\dim \ker(X_j|_{P_0 \mathcal{H}}) = k - q, \quad (4.3)$$

the same finite number for $j = 1, 2$. The two residual positive compact operators have the same nonzero eigenvalues with multiplicity and the same kernel dimension, so the spectral theorem makes them unitarily equivalent. If $P_0 = 0$, the existence of X_j simply forces the residual tail to be empty, and the same conclusion holds.

Choose a conjugating unitary on every nontrivial spectral block, take their orthogonal direct sum, and use the identity where both X_j vanish. The resulting unitary V conjugates X_1 to X_2 and commutes with H , since each nontrivial summand lies within an eigenspace of H . $\square$

Lemma 5 (Occupied supporting faces are filled). *For $A \in \mathcal{B}(\mathcal{H})$, the intersection of $W_C(A)$ with any supporting line of $K = \overline{W_C(A)}$ is convex.*

Proof. After multiplying A by a scalar of modulus one, take the supporting line to be vertical. Write $H = \operatorname{Re} A$ and $G = \operatorname{Im} A$. Two points of $W_C(A)$ on this line have the form

$$\operatorname{Tr}(X_j A) = \alpha + i\beta_j,$$

where $X_j \in \mathcal{U}(C)$ maximize $\operatorname{Tr}(X H)$. By Lemma 4, $X_2 = V X_1 V^*$ for a unitary V commuting with H .

Choose a bounded selfadjoint Borel-functional-calculus logarithm L of V . Because V commutes with H , so does L . For $V_t = e^{itL}$ and $X_t = V_t X_1 V_t^*$, every X_t lies in $\mathcal{U}(C)$ and

$$\operatorname{Tr}(X_t H) = \operatorname{Tr}(X_1 H) = \alpha.$$

The real-valued function $t \mapsto \operatorname{Tr}(X_t G)$ is continuous and has endpoint values β_1, β_2 . It attains every intermediate value, so the entire segment between the two original points lies in $W_C(A)$. $\square$

5 Proof of Theorem 1

Put $S = W_C(A)$ and $K = \overline{S}$. The set K is convex by [2, Theorem 3.4 and Corollary 4.6]. The unitary group is norm path-connected: if U is unitary, a bounded selfadjoint logarithm produces the path $e^{it \log U}$. Hence S , a continuous image of the unitary group, is path-connected.

If K has affine dimension at most one, S is a connected subset of a line (or a point), hence an interval and therefore convex. Suppose K has nonempty planar interior. Take $x, y \in S$ and $z = (1 - t)x + ty$ with $0 < t < 1$. If $z \in \text{int } K$, Lemma 3 gives $z \in S$. If $z \in \partial K$, choose a supporting real-linear functional ϕ at z . Since $\phi \leq \phi(z)$ on K and

$$\phi(z) = (1 - t)\phi(x) + t\phi(y),$$

both inequalities must be equalities. Thus x and y lie on the same supporting line, and Lemma 5 gives $z \in S$. Every segment with endpoints in S is contained in S , proving the theorem. $\square$

6 Verification, limitations, and novelty

- The common-tail construction was checked at the operator level: the leading spectral truncations agree exactly, while the two discarded tails give the explicit $2\|C - C_N\|_1$ bound. The finite-dimensional convex subset in (3.1) consists of genuine members of the original unitary orbit.
- The only place where $\dim \ker C < \infty$ is used is (4.2)–(4.3). It prevents kernel multiplicity from moving invisibly between the threshold space and its complement.
- The proof does not claim closedness. It fills the planar interior and each supporting face that already contains ordinary orbit points.
- When $\dim \ker C = \infty$ and C has infinite rank, cardinal subtraction does not determine the residual kernel multiplicity in (4.3). The commutant-orbit lemma is therefore not proved by this argument. The general selfadjoint problem also remains open here.
- A bounded novelty search on 22 July 2026 used arXiv/web queries for “ C -numerical range convex”, “positive trace class”, “finite-dimensional kernel”, and the Dirr–vom Ende open problem. It found arXiv:1712.01023, 1808.06898, 2009.01300, and the orbit-closed follow-up arXiv:2106.11205, but no later theorem with this finite-kernel conclusion and arbitrary bounded A . This is evidence, not a definitive literature certification.

Human-review recommendation. Verify and retain as a substantial partial theorem. The main points to audit are the simultaneous common-tail construction in Lemma 2 and the finite residual-kernel count in (4.3) when Proposition 5.12 of [2] is applied to ordinary maximizers.

References

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